PROBLEM 40

Spring-Coupled Particles on a Vertical Axis and a Revolving Ring

Original statement · Problems.pdf, p. 28

Two Particles in a Rotating Frame

Two heavy particles P and Q, each of mass m, are connected with a spring of stiffness k as shown in the figure. An inertial reference frame S_0 is defined with origin O and vector basis $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$. Particle P slides without friction along the vertical axis, while particle Q slides without friction on a ring of radius R contained in the horizontal plane defined by O, $\mathbf{i}_0$ and $\mathbf{j}_0$. The center D of this ring is located a distance ℓ from O and rotates around the vertical axis of S_0 with a known constant angular rate Ω. A rotating reference frame S_1 is defined so that the $\mathbf{j}_1$ vector follows the position of D. We define the generalized coordinates ζ (vertical position of P) and ϕ (angle of Q on the circle from $\mathbf{j}_1$) as shown.

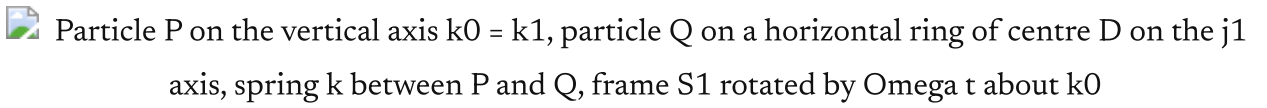


Figure: Particle P on the vertical axis k0 = k1, particle Q on a horizontal ring of centre D on the j1 axis, spring k between P and Q, frame S1 rotated by Omega t about k0 (Problems.pdf, p. 28).

(a) How many degrees of freedom does the system have? Define the vectors $\mathbf{i}_1, \mathbf{j}_1$ and $\mathbf{k}_1$ in terms of the vectors of $\mathcal{B}_0$.

(b) Write down the general expression of the position, velocity, and acceleration of P and Q with respect to S_1 in terms of ζ, ϕ and their derivatives.

(c) Identify and write down the general expression of all the forces acting on P and Q.

(d) Using S_1 for your analysis, write down the scalar equations of motion of P and Q.

(e) Starting from these equations, write down a system of two differential equations for ζ and ϕ, that do not depend on any other unknowns of the problem.

(f) Find one conservation law of the problem (i.e., a first integral of motion) in reference frame S_1.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** two heavy particles of mass m. P slides on the fixed vertical axis Oz_0; Q slides on a horizontal ring of radius R in the plane $z_0 = 0$ whose centre D, at distance ℓ from O, revolves about Oz_0 at constant rate Ω. A spring of stiffness k joins P and Q.
- **Hypotheses:**
  1. $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial, $\mathbf{k}_0$ vertical upward, gravity $-g\,\mathbf{k}_0$.
  2. Both contacts are smooth and bilateral: the reactions have no component along the direction in which each particle can move.
  3. The spring is linear and massless, with zero natural length (no natural length is given), so the force it exerts on P is $k\left(\mathbf{r}^Q - \mathbf{r}^P\right)$ and on Q the opposite.
  4. The ring moves rigidly with S_1 (only its centre's motion is prescribed; it does not spin relative to S_1, which is irrelevant anyway for a smooth circular wire).
  5. ϕ is measured from $\mathbf{j}_1$ in the positive sense about $\mathbf{k}_1$ (from $\mathbf{j}_1$ towards $-\mathbf{i}_1$), as in the figure; this is also the convention of the official key, whose ring reaction contains the radial vector $-\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1$.
- **Why S_1:** in S_1 the ring is fixed (centre at $\ell\,\mathbf{j}_1$), so Q moves on a fixed circle described by ϕ alone and P stays on the axis. The inertia forces (Notes Eq. 4.62) are the price; for P, on the rotation axis, they vanish.

Symbol	Meaning	SI unit
m	mass of each particle	kg
k	spring stiffness	N/m
R	radius of the ring	m

Symbol	Meaning	SI unit
ℓ	distance OD	m
Ω	constant angular rate of D	rad/s
g	gravitational acceleration	m/s^2
ζ, ϕ	height of P, angle of Q on the ring	m, rad
$N_1, \dots, N_4$	reaction components	N

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Frames:** $S_0 = \{O; \mathcal{B}_0\}$ inertial; $S_1 = \{O; \mathcal{B}_1\}$ with the same origin, $\mathbf{k}_1 = \mathbf{k}_0$ and $\mathbf{j}_1$ pointing from O to D. S_1 is rotated by Ωt about $\mathbf{k}_0$ (the figure measures Ωt from $\mathbf{i}_0$ to $\mathbf{i}_1$), and we take $S_1 \equiv S_0$ at $t = 0$:

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i}_0 + \sin\Omega t\,\mathbf{j}_0, \qquad \mathbf{j}_1 = -\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0, \qquad \mathbf{k}_1 = \mathbf{k}_0$$

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & -\sin\Omega t & 0 \\ \sin\Omega t & \cos\Omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\Omega t + \sin^2\Omega t = 1$$

- **Orthogonality:** each row has norm $\cos^2\Omega t + \sin^2\Omega t = 1$ (or 1), and the first two rows have dot product $\cos\Omega t\sin\Omega t - \sin\Omega t\cos\Omega t = 0$, so $[{}_0R_1][{}_0R_1]^T = I$.
- **Motion of S_1:** $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$; simple rotation about $\mathbf{k}_0$ (Slide 16): $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_1$, and $\boldsymbol{\alpha}_{10} = \mathbf{0}$ because Ω and $\mathbf{k}_0$ are constant.
- **Ring basis** (attached to Q, defined in $\mathcal{B}_1$):

$$\mathbf{e}_r = -\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1, \qquad \mathbf{e}_\phi = \frac{d\mathbf{e}_r}{d\phi} = -\cos\phi\,\mathbf{i}_1 - \sin\phi\,\mathbf{j}_1, \qquad \mathbf{k}_1$$

$\mathbf{e}_r$ points from D to Q and $\mathbf{e}_\phi$ is tangent to the ring. Check: $\mathbf{e}_r \times \mathbf{e}_\phi = [(-\sin\phi)(-\sin\phi) - (\cos\phi)(-\cos\phi)]\,\mathbf{k}_1 = \mathbf{k}_1$, so $\{\mathbf{e}_r, \mathbf{e}_\phi, \mathbf{k}_1\}$ is right-handed and orthonormal. The inverse relations used below are $\mathbf{j}_1 = \cos\phi\,\mathbf{e}_r - \sin\phi\,\mathbf{e}_\phi$ and $\mathbf{i}_1 =$

$-\sin\phi\,\mathbf{e}_r - \cos\phi\,\mathbf{e}_\phi$.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Degrees of freedom and the basis $\mathcal{B}_1$

Pedagogical Justification: count coordinates minus independent constraints for each particle.

P is free in space (3 coordinates) but constrained to the vertical axis ($x = y = 0$): 1 DOF, the height ζ. Q (3 coordinates) is constrained to the plane $z_1 = 0$ and to the circle $\left|\mathbf{r}^Q - \overrightarrow{OD}\right| = R$: 1 DOF, the angle ϕ. The motion of D is prescribed, so it adds no unknown. Hence

$$\text{DOF} = 6 - 4 = 2, \qquad \text{generalized coordinates } (\zeta, \phi)$$

The vectors of $\mathcal{B}_1$ in terms of $\mathcal{B}_0$ are those given in Phase 2.

(b) Relative kinematics of P and Q in S_1

Pedagogical Justification: O is the origin of S_1, so $\mathbf{r}_1 = \overrightarrow{OP}$ or $\overrightarrow{OQ}$ (Notes Eq. 3.16). The vectors $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ are constant in S_1; $\mathbf{e}_r$ and $\mathbf{e}_\phi$ change only through $\phi(t)$, so the chain rule gives $\left.\dfrac{d\mathbf{e}_r}{dt}\right|_1 = \dfrac{d\mathbf{e}_r}{d\phi}\dot\phi = \dot\phi\,\mathbf{e}_\phi$ and $\left.\dfrac{d\mathbf{e}_\phi}{dt}\right|_1 = \dfrac{d\mathbf{e}_\phi}{d\phi}\dot\phi = \dot\phi\,(\sin\phi\,\mathbf{i}_1 - \cos\phi\,\mathbf{j}_1) = -\dot\phi\,\mathbf{e}_r$.

Particle P:

$$\mathbf{r}_1^P = \zeta\,\mathbf{k}_1, \qquad \mathbf{v}_1^P = \dot\zeta\,\mathbf{k}_1, \qquad \mathbf{a}_1^P = \ddot\zeta\,\mathbf{k}_1$$

Particle Q: $\overrightarrow{OQ} = \overrightarrow{OD} + \overrightarrow{DQ}$ with $\overrightarrow{OD} = \ell\,\mathbf{j}_1$ (constant in S_1):

$$\mathbf{r}_1^Q = \ell\,\mathbf{j}_1 + R\,\mathbf{e}_r = -R\sin\phi\,\mathbf{i}_1 + (\ell + R\cos\phi)\,\mathbf{j}_1$$

$$\mathbf{v}_1^Q = R\left.\frac{d\mathbf{e}_r}{dt}\right|_1 = R\dot\phi\,\mathbf{e}_\phi = -R\dot\phi\cos\phi\,\mathbf{i}_1 - R\dot\phi\sin\phi\,\mathbf{j}_1$$

By the product rule, $\mathbf{a}_1^Q = R\ddot\phi\,\mathbf{e}_\phi + R\dot\phi\left.\frac{d\mathbf{e}_\phi}{dt}\right|_1$:

$$\mathbf{a}_1^Q = R\ddot\phi\,\mathbf{e}_\phi - R\dot\phi^2\,\mathbf{e}_r$$

(c) Forces on P and Q (actual and inertia forces in S_1)

> **Pedagogical Justification:** in S_1 we apply $m\mathbf{a}_1 = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63) with $\mathbf{F}_i$ from Notes Eq. 4.62. Here $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$, so only the centrifugal and Coriolis forces survive (Slide 28).

Spring. The relative position is

$$\mathbf{r}^Q - \mathbf{r}^P = \ell\,\mathbf{j}_1 + R\,\mathbf{e}_r - \zeta\,\mathbf{k}_1 = -R\sin\phi\,\mathbf{i}_1 + (\ell + R\cos\phi)\,\mathbf{j}_1 - \zeta\,\mathbf{k}_1$$

so the spring force on P is $\mathbf{F}_s^P = k\left(\mathbf{r}^Q - \mathbf{r}^P\right)$ and on Q it is $\mathbf{F}_s^Q = -k\left(\mathbf{r}^Q - \mathbf{r}^P\right)$. In the ring basis, using $\mathbf{j}_1 = \cos\phi\,\mathbf{e}_r - \sin\phi\,\mathbf{e}_\phi$:

$$\mathbf{F}_s^Q = -k\left(\ell\cos\phi + R\right)\mathbf{e}_r + k\ell\sin\phi\,\mathbf{e}_\phi + k\zeta\,\mathbf{k}_1$$

Forces on P:

- Weight $-mg\,\mathbf{k}_1$.
- Reaction of the smooth axis, perpendicular to $\mathbf{k}_1$: $\mathbf{N}_P = N_1\,\mathbf{i}_1 + N_2\,\mathbf{j}_1$.
- Spring $\mathbf{F}_s^P = k\left[-R\sin\phi\,\mathbf{i}_1 + (\ell + R\cos\phi)\,\mathbf{j}_1 - \zeta\,\mathbf{k}_1\right]$.
- Inertia forces: P is on the rotation axis. $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = \Omega\zeta\,\mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$, so $\mathbf{F}_C^P = \mathbf{0}$; $\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P = \Omega\dot\zeta\,\mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$, so $\mathbf{F}_{COR}^P = \mathbf{0}$.

Forces on Q:

- Weight $-mg\,\mathbf{k}_1$.

- Reaction of the smooth ring, perpendicular to the tangent $\mathbf{e}_\phi$: $\mathbf{N}_Q = N_3\,\mathbf{e}_r + N_4\,\mathbf{k}_1 = N_3\left(-\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1\right) + N_4\,\mathbf{k}_1$.
- Spring $\mathbf{F}_s^Q$ above.
- Centrifugal force. Since $\mathbf{r}_1^Q$ has no $\mathbf{k}_1$ component, $\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^Q\right) = -\Omega^2 \mathbf{r}_1^Q$ (from $\mathbf{k}_1 \times (\mathbf{k}_1 \times \mathbf{i}_1) = -\mathbf{i}_1$ and $\mathbf{k}_1 \times (\mathbf{k}_1 \times \mathbf{j}_1) = -\mathbf{j}_1$), hence

$$\mathbf{F}_C^Q = m\Omega^2 \mathbf{r}_1^Q = m\Omega^2\left(\ell\,\mathbf{j}_1 + R\,\mathbf{e}_r\right) = m\Omega^2\left(\ell\cos\phi + R\right)\mathbf{e}_r - m\Omega^2 \ell \sin\phi\,\mathbf{e}_\phi$$

- Coriolis force. From $\mathbf{e}_r \times \mathbf{e}_\phi = \mathbf{k}_1$ (cyclic: $\mathbf{e}_\phi \times \mathbf{k}_1 = \mathbf{e}_r$), $\mathbf{k}_1 \times \mathbf{e}_\phi = -\mathbf{e}_r$:

$$\mathbf{F}_{COR}^Q = -2m\Omega\,\mathbf{k}_1 \times R\dot{\phi}\,\mathbf{e}_\phi = 2m\Omega R\dot{\phi}\,\mathbf{e}_r$$

- Euler force and force due to the origin's acceleration: zero for both particles.

(d) Scalar equations of motion in S_1

> **Pedagogical Justification:** project $m\mathbf{a}_1 = \mathbf{F} + \mathbf{F}_i$ for each particle on an orthonormal basis adapted to its constraint: $\{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ for P and $\{\mathbf{e}_r, \mathbf{e}_\phi, \mathbf{k}_1\}$ for Q. This gives $3 + 3 = 6$ scalar equations for the 6 unknowns $\zeta, \phi, N_1, N_2, N_3, N_4$.

Particle P ($m\ddot{\zeta}\,\mathbf{k}_1 = -mg\,\mathbf{k}_1 + N_1\mathbf{i}_1 + N_2\mathbf{j}_1 + \mathbf{F}_s^P$):

$$\mathbf{i}_1: \quad 0 = N_1 - kR\sin\phi$$

$$\mathbf{j}_1: \quad 0 = N_2 + k\left(\ell + R\cos\phi\right)$$

$$\mathbf{k}_1: \quad m\ddot{\zeta} = -mg - k\zeta$$

Particle Q ($m\left(R\ddot{\phi}\,\mathbf{e}_\phi - R\dot{\phi}^2\mathbf{e}_r\right) = -mg\,\mathbf{k}_1 + N_3\mathbf{e}_r + N_4\mathbf{k}_1 + \mathbf{F}_s^Q + \mathbf{F}_C^Q + \mathbf{F}_{COR}^Q$):

$$\mathbf{e}_r: \quad -mR\dot{\phi}^2 = N_3 - k\left(\ell\cos\phi + R\right) + m\Omega^2\left(\ell\cos\phi + R\right) + 2m\Omega R\dot{\phi}$$

$$\mathbf{e}_\phi: \quad mR\ddot{\phi} = k\ell\sin\phi - m\Omega^2\ell\sin\phi$$

$$\mathbf{k}_1: \quad 0 = -mg + N_4 + k\zeta$$

(e) Two equations free of the reactions

> **Pedagogical Justification:** the reactions appear only in the projections perpendicular to the allowed motions; the projections along the allowed motions ($\mathbf{k}_1$ for P, $\mathbf{e}_\phi$ for Q) are free of them.

Dividing the $\mathbf{k}_1$ equation of P and the $\mathbf{e}_\phi$ equation of Q by m and mR:

$$\boxed{\ddot{\zeta} + \frac{k}{m}\zeta = -g, \qquad \ddot{\phi} = \frac{\ell}{R}\left(\frac{k}{m} - \Omega^2\right)\sin\phi}$$

The other four equations then give the reactions once $\zeta(t), \phi(t)$ are known: $N_1 = kR\sin\phi$, $N_2 = -k\left(\ell + R\cos\phi\right)$, $N_3 = -mR\dot{\phi}^2 + \left(k - m\Omega^2\right)\left(R + \ell\cos\phi\right) - 2m\Omega R\dot{\phi}$, $N_4 = mg - k\zeta$. In particular the ring reaction has the form $\mathbf{N}_Q = N_3\left(-\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1\right) + N_4\,\mathbf{k}_1$ of the official key.

The two equations are **decoupled**: the vertical spring component acts only on P's coordinate, and the horizontal one only on Q's, because the spring force is linear in $\mathbf{r}^Q - \mathbf{r}^P$ and P moves along $\mathbf{k}_1$ while Q moves in the plane $\perp \mathbf{k}_1$.

(f) A conservation law in S_1

> **Pedagogical Justification:** in S_1 the constraints (axis and ring) are fixed and smooth, so the reactions do no work; the Coriolis force on Q is perpendicular to $\mathbf{v}_1^Q$ (Slide 31); the weight and spring are conservative; and with $\boldsymbol{\omega}_{10}$ constant the centrifugal force has potential $-\frac{1}{2}m\Omega^2 r_\perp^2$ (Slide 30). Hence the mechanical energy of the system in S_1 is conserved (Slide 29).

Potential energies in S_1: weight of P, $mg\zeta$ (weight of Q: $mg\cdot 0$); spring $\frac{1}{2}k\left|\mathbf{r}^Q-\mathbf{r}^P\right|^2$; centrifugal on Q, $-\frac{1}{2}m\Omega^2\left|\mathbf{r}_1^Q\right|^2$ (here $r_\perp^Q=|\mathbf{r}_1^Q|$). With

$$\left|\mathbf{r}^Q-\mathbf{r}^P\right|^2=R^2\sin^2\phi+(\ell+R\cos\phi)^2+\zeta^2=\ell^2+R^2+2\ell R\cos\phi+\zeta^2,\qquad \left|\mathbf{r}_1^Q\right|^2=\ell^2+R^2+2\ell R\cos\phi$$

and dropping the constant $\frac{1}{2}\left(k-m\Omega^2\right)\left(\ell^2+R^2\right)$:

$$\boxed{E_1=\frac{1}{2}m\left(\dot\zeta^2+R^2\dot\phi^2\right)+mg\zeta+\frac{1}{2}k\zeta^2+\left(k-m\Omega^2\right)\ell R\cos\phi=\text{const}}$$

Check with the equations of part (e) (chain rule $\frac{d}{dt}\cos\phi=-\dot\phi\sin\phi$, $\frac{d}{dt}\zeta^2=2\zeta\dot\zeta$):

$$\frac{dE_1}{dt}=m\dot\zeta\ddot\zeta+mR^2\dot\phi\ddot\phi+mg\dot\zeta+k\zeta\dot\zeta-\left(k-m\Omega^2\right)\ell R\dot\phi\sin\phi=m\dot\zeta\left(\ddot\zeta+g+\frac{k}{m}\zeta\right)+mR\dot\phi\left(R\ddot\phi-\left(\frac{k}{m}-\Omega^2\right)\ell\sin\phi\right)=0$$

Because the equations are decoupled, each bracket vanishes separately, so in fact two first integrals exist: $\frac{1}{2}m\dot\zeta^2+mg\zeta+\frac{1}{2}k\zeta^2=\text{const}$ and $\frac{1}{2}mR^2\dot\phi^2+\left(k-m\Omega^2\right)\ell R\cos\phi=\text{const}$. The constants are fixed by initial conditions, which the statement does not give.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Particle P:** $\ddot\zeta+(k/m)\zeta=-g$ is a harmonic oscillator about $\zeta_{eq}=-mg/k$ with angular frequency $\sqrt{k/m}$, unaffected by the rotation (it lies on the axis, where centrifugal and Coriolis forces vanish).

- **Particle Q:** $\ddot{\phi} = \frac{\ell}{R}\left(\frac{k}{m} - \Omega^2\right)\sin\phi$ is a pendulum equation. The spring pulls Q towards the axis (towards $\phi = \pi$, the point of the ring nearest O) and the centrifugal force pushes it outward (towards $\phi = 0$). If $\Omega^2 < k/m$ the stable position is $\phi = \pi$; if $\Omega^2 > k/m$ it is $\phi = 0$; at $\Omega^2 = k/m$ the two effects cancel exactly and $\dot{\phi}$ is constant. The Coriolis force only loads the ring radially (N_3).
- **Limits:** $\Omega \to 0$ gives a fixed ring and $\ddot{\phi} = \frac{k\ell}{mR}\sin\phi$; $\ell \to 0$ (ring centred on the axis) gives $\ddot{\phi} = 0$, uniform motion, since every point of the ring is then equidistant from the axis.
- **Dimensions:** $[k/m] = \mathrm{N/(m\,kg)} = \mathrm{s^{-2}}$, so $[(k/m)\zeta] = \mathrm{m/s^2} = [g] = [\ddot{\zeta}]$; $[(\ell/R)(k/m - \Omega^2)] = \mathrm{s^{-2}} = [\ddot{\phi}]$; $[N_3]$: $[mR\dot{\phi}^2] = [kR] = [m\Omega R\dot{\phi}] = \mathrm{N}$; $[E_1] = \mathrm{kg\,m^2/s^2} = [k\zeta^2] = [k\ell R] = \mathrm{J}$.
- **Official key** (Problems.pdf, p. 76): "One of the forces acting on Q is the reaction of the ring, $\mathbf{N}_Q = N_3(-\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1) + N_4\,\mathbf{k}_1$", which matches part (c).
- **Symbolic check (sympy):** $\mathbf{a}_0$ of both particles obtained by differentiating $[{}_0R_1]\,\mathbf{r}_1$ twice matches Notes Eq. 3.27; the six scalar equations, the four reactions, and $dE_1/dt = 0$ along the motion are confirmed.

Source: Problems.pdf (PDF page 28); key on PDF page 76